CHAPTER 3

Role of the gallbladder in our metabolism and immune system

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Introduction

Bile acid production, secretion and storage

Bile is a complex physiological aqueous secretion that contains bile salts, phospholipids, amino acids, cholesterol, enzymes, porphyrins, vitamins, steroids, and heavy metals (Boyer, 2013). The original site of bile formation is the liver, and bile-secreting cells are called hepatocytes, which are later modified by cholangiocytes (Lazaridis et al., 2004). The bile duct epithelium is the modification site of bile, and some carrier proteins are present there that help in the transportation of bile. These carrier proteins have been found to create a concentration gradient that makes the water flow through osmosis, resulting in the active secretion of solutes (Boyer, 2013). The addition of bile salt influences the rate of bile flow (Vonk et al., 1979). Bile is stored in the gallbladder, and the gallbladder starts emptying when the digestion process starts in the upper GIT and when the food bolus goes down into the intestine (Golding & Wooster, 2010). Intestinal fat absorption is generally facilitated by bile salt emulsification of the fat present in food, and cholesterol is eliminated from the body through biliary excretion (Macierzanka et al., 2019; Trautwein et al., 2003).

Bile acid action in immunity and metabolism

The detergent nature of bile acid enhances the immunological nature of bile acid, and some bile acid receptors have anti-inflammatory activity. On the other hand, bile acids maintain metabolism in our body by maintaining normal energy homeostasis, normal blood glucose levels, and preventing atherosclerosis.

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Physiology of gallbladder

The human gallbladder is basically a pyramidal sac-like structure, closely attached to the liver with the help of soft connective tissues, blood vessels, and certain lymphatic vessels (Paul & Bhatia, 2020). There are three parts of the gallbladder that have been distinct, i.e., neck, corpus, and fundus (Pasternak et al., 2013). The blood supply to the gallbladder occurs through the cystic artery, originating from hepatic arteries or duodenal arteries, and the venous blood flows from the gallbladder to the system of the small veins, which is emptied out into the portal veins (Halvorsen & Myking, 1971). The mucosal wall of the gallbladder consists of a muscle layer and serosa (Nagahashi et al., 2007).

Epithelial cells

A layer of columnar epithelial cells is present inside the mucosal layer. These columnar epithelial cells are also called principal cells. The surface epithelium consists of three types of cells: (1) columnar clear cells; (2) dark tuft osmophilic cells; and (3) basal clear cells (Gilloteaux et al., 1989). On the inner surface, the epithelial cells of the gallbladder are often exposed to a high levels and fluctuating concentrations of biliary cholesterol, and later this cholesterol is absorbed by the cells either actively or passively; the process is bidirectional (Tauscher & Kuver, 2003). Mucin secretion happens in epithelium cells (Auth et al., 1993).

Lamina propria

In between epithelial cells and muscular layers, this lamina propria is present, which occurs to pack the mucosal folds. Loose connective tissues, plasma cells, lymphocytes, nerve fibers, small blood vessels, and macrophages are the main components forming the lamina propria. Neutrophils and eosinophils are usually absent (Elfing, 1960).

Muscle layer

Usually, the random arrangement of smooth muscle cellular bundles constitutes this muscular layer of the gallbladder. Tunica muscularis is the single muscle layer present in the gallbladder. The muscle cells are highly porous, so invagination from the mucosal layer happens with ease. These mucosal invaginations are highly prone to inflammation. The peri-muscular layer is formed with some fibroelastic and adipose tissues (Elfing, 1960).

The core function of the gallbladder is to store the biliary materials formed and secreted by the liver. As the water and inorganic materials present within the biliary compounds are pulled out, the bile becomes concentrated, and the secretion of mucus turns the bile into a viscous form (Center, 2009). The motility function of the gallbladder is maintained by two main factors: neural and endocrine (Wu et al., 2013). Both sympathetic and parasympathetic nervous innervation are seen inside the gallbladder, where gallbladder motility is significantly influenced by cephalic regulation. However, the sympathetic innervation of the gallbladder has an inhibitory effect on its motility (Jazrawi, 2003). Gastrointestinal hormones and peptides also act as major regulating factors in gallbladder motility. Cholecystokinin, or CCK, Gastrin, Secretin, Motilin, Neuropeptide Y, Substance P, Gastrin-releasing peptide, and Histamine are the prokinetic agents that act on gallbladder motility (Tierney et al., 1993).

Formation, storage, and secretion of bile

Formation of bile acid

The precursor of bile acid synthesis is cholesterol, and 17 enzymes act as precursors of this bile acid synthesis process. The initial outcome is called primary bile acid. In the case of humans, chenodeoxycholic acid is termed primary bile acid. Further, the chemical composition of bile acid and the bile acid pool is controlled by the gut microbiome. This gut microbiome is further responsible for the conversion to secondary and tertiary bile acids (Hylemon & Harder, 1998).

Classical or formal pathway

There are several pathways that control the formation of bile acid. However, in the classical pathway, the primary component, cholesterol, is converted into 7α-hydroxycholesterol with the help of a microsomal cytochrome P450 enzyme named cholesterol 7α-hydroxylase, manifested only in the liver. This cytochrome 450 enzyme is the only rate-limiting enzyme for bile acid formation and is present in the microsomal and mitochondrial compartments of cells. The hydroxylation of the side chains of cholesterol occurs later on, and 4-hydroxycholesterol, 25-hydroxycholesterol, and 27-hydroxycholesterol are formed. In the case of humans, the mRNA protein associated with the 24-hydroxylase enzyme is detected in normal brain cells. This 24-hydroxylase enzyme is

responsible for bile acid synthesis in most of the phases. In contrast, it has a crucial role in the turnover of cholesterol in the brain. On the other hand, the 25-hydroxylase enzyme is not a P450 enzyme but can catalyze lipid metabolism, and it uses up oxygen and acts on cholesterol catabolism as enzyme 24-hydroxylase does. 25-hydroxycholesterol and 27-hydroxycholesterol are converted to bile acid intermediates, and CYP7B1 oxysterol 7α-hydroxylase is the enzyme that is responsible for the whole process (Russell, 2003).

Alternate or acidic pathway

The initiation of the acidic pathway enhances the oxysterols, which act on the primary activation of liver X receptors, or LXRs. Reverse transport of cholesterol occurs by the activation of LXR with the help of HDL, followed by reduced peripheral cholesterol, and simultaneously, the induction of cholesterol transporters like ATP-binding transporters (ABCA1, ABCG5, and ABCG8) and ApoE occurs (Hong & Tontonoz, 2014). Similarly, the activation of LXR also results in hepatic lipogenesis by activating the expression of sterol regulatory element binding protein-1c (SREBP-1c), fatty acid synthase, acetyl-CoA carboxylase 1, and stearoyl-CoA desaturase 1 (Joseph et al., 2002; Repa et al., 2000; Talukdar & Hillgartner, 2006). Later, oxysterols bind to proteins and maintain cholesterol homeostasis. Insulin-induced gene protein (INSIG) helps in the maturation of sterol regulatory element binding protein (SERBP) (Radhakrishnan et al., 2007). Lipid biosynthesis is increased by the interaction of INSIG and SERBP cleavage-activating proteins. LDL-derived cholesterol is mobilized by Niemann-Pick type C protein 1 (NPC1). Thus, the endosomal cholesterol is cleared (Higgins et al., 1999) (Fig. 3.1).

Enterohepatic circulation of bile acids

As we know, the circulating and *de novo* synthesized bile acid is termed primary bile acid, which is formed from cholesterol by passing through a lot of steps (Hofmann, 2009). In the case of man, cholic acid and chenodeoxycholic acid are called primary bile acids. Later, this primary bile acid is converted into secondary bile aid with the help of intestinal microorganisms, and this modification occurs at the distal part of the intestine (Floch, 2002). While passing through the intestine, bile acid plays a key role in the absorption of lipids and fat-soluble vitamins. According to the

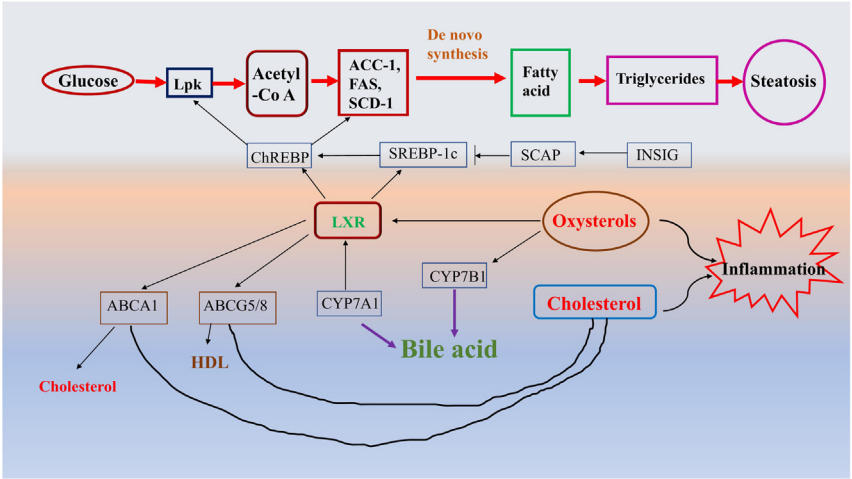

Figure 3.1 The acidic pathway of bile acid synthesis and the regulation of metabolism.

studies, intestinal enterocytes help in the reabsorption of 95% of bile acid, and the rest 5% gets excreted out *via* stool and causes a deficit. Later, this deficit of bile acid is made up by the next *de novo* synthesis. About 4–12 times the recycling of bile acids happens in between the hepatocytes of the liver and intestinal enterocytes, and this circulation is called the enterohepatic circulation of bile acids (Mok et al., 1977). Bile composition is determined by the size of bile acid components and the bile acid pool (Garbutt & Kenney, 1972). The high concentration of hepatic bile acid causes the toxicity in the hepatic system, and that is why hepatocytes can promote bile acid excretion to reduce this toxicity (Zollner et al., 2006). The low plasma bile acid concentration is caused by the hepatic clearance of bile acid, and virtual renal excretion is also absent. The excretion of bile occurs through fecal excretion (Hofmann, 2009).

Bile acid and blood-brain barrier (BBB)

Internal carotids and vertebral arteries carry the bile acids to the brain, and these arteries together form a ring—the circle of Willis (Abbott et al., 2010; Bernacki et al., 2008). It is proven that conjugated bile acids as well as unconjugated bile acids can cross the BBB. Contradictorily, the unconjugated bile acids are diffused across the BBB, as chenodeoxycholic acid, cholic acid, and cholic acid are subjected to being diffused out throughout the phospholipid bilayer. But conjugated bile acids are large in structure

and amphipathic in nature, so they cross the BBB with the help of active transporters. As well as the liver, intestine, and kidney, some xenobiotic transporters are also present in the BBB. Those transporters act as protectors of the brain as they carry toxic substances from the brain to the bloodstream (Mertens et al., 2017).

Role of gallbladder and biliary tree in immunity

In the anatomical description, the biliary tree has been divided into intrahepatic and extrahepatic segments, and gallbladder epithelium is included within that, and those epithelial cells have a frontier role in the innate and adaptive immune systems. The adaptive immunity includes B cell functions carried out by immunoglobulins, and bile carries a minute concentration of Igs because the plasma cell membrane of the biliary epithelium secretes a little concentration of IgA further, which is secreted into the bile, resulting in the inhibition of pathogens (Nault et al., 2013; Reynoso-Paz et al., 1999). Bile salts get retrieved in the distal part of the small intestine, followed by secretion into the portal circulation, and later on, they are excreted out by the liver (Fig. 3.2). However, bile salt transporters play a pivotal role in the enterohepatic circulation of bile. The physiological activities of bile salt are mediated through some receptors like plasma membrane receptors (e.g., TGR5), some nuclear bile salt receptors (e.g., FXR), and a lot of cell surface receptors that get activated

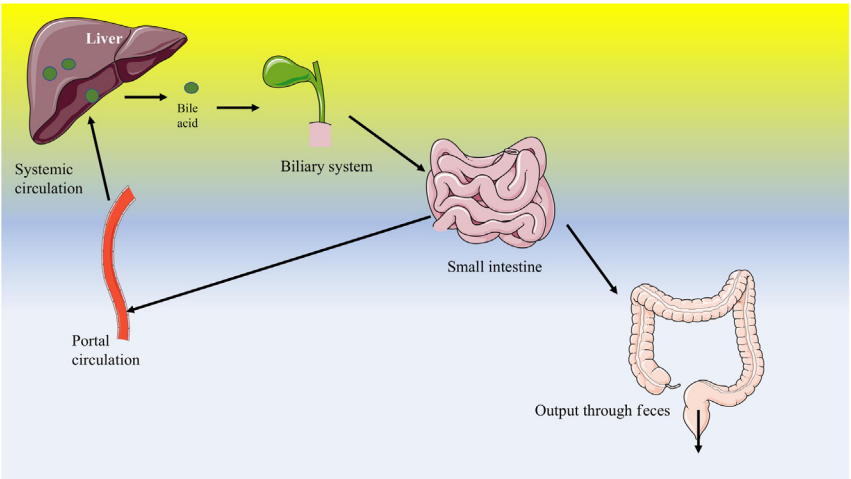

Figure 3.2 Schematic diagram of the enterohepatic circulation of bile acid.

in a direct (e.g., S1PR2 activation by taurine-conjugated CA 12) or passive manner (e.g., FGFR4 activation by bile salt-induced FGF1913), and also there are some bile salt sensing proteins present (Fiorucci et al., 2021). Chemically, bile salt is a biological detergent, has direct antimicrobial properties, and is responsible for maintaining gut microbiota. It generally acts as the mechanism of disruption of membrane integrity (Fig. 3.3). There are other factors that also control the gut microbiota, like age, lifestyle, genetic factors, and diet, and the microbiome content varies from person to person in a wide range. This variation is dependent upon the metabolic and immunological interactions with the host (Candela et al., 2015). This detergent activity of bile salt has a wide connection with the log value of hydrophobicity K_{OW}, which is demonstrated by bile salt toxicity. Here, a negative correlation of conjugated moiety and a number of hydroxyl groups occurs, either glycine or taurine, located at the bile acid carboxyl group site (Begley et al., 2005). According to the ranking of detergent activity, the most toxic bile salt is lithocholic acid, which contains a hydroxyl group (Heuman, 1989). The second pathway of toxic activity against bacteria happens when the bile salt gets deconjugated and free bile salt invades the cells of bacteria, and this mechanism occurs by flip-flop action when it is protonated in nature (Kurdi et al., 2000, 2003). After entering the bacterial cell, bile acid gets deprotonated because the internal environment of a bacterial cell is alkaline in nature. According to some previous evidence, *Bifidobacteria* and *Lactobacillus* have shown reduced

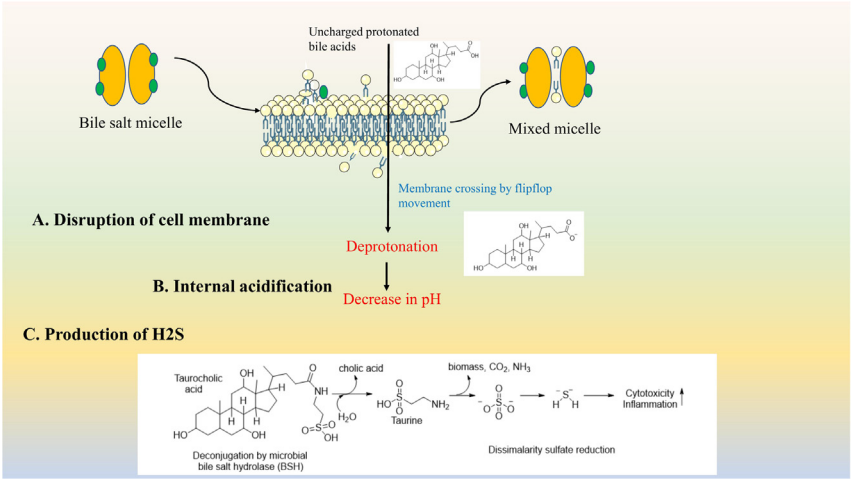

Figure 3.3 Mechanisms of action of toxicity caused by bile salt.

internal pH due to the high concentration of bile salts within the cell (Kurdi et al., 2006). The ATP generation decreases because of the disrupted pH gradient inside the bacterial cell as a result of the drastic reduction in pH. Till now, some reasons causing protein degradation as a result of reduced pH inside the cells are unknown. There is another pathway based on the toxicity of bile salt, which demonstrates that taurine-conjugated bile salt gets metabolized and results in the formation of taurine and some residues of hydrogen sulfide (H_2S) (Ridlon et al., 2016). Catalysts occurring in the deconjugation process are bile salt hydrolases. These enzymes are basically found in a lot of gram-positive bacteria like *Lactobacillus*, *Enterococcus*, *Clostridium*, and *Bifidobacterium*, although there are some Bacteroides present that also show bile salt hydrolase activity. Later, taurine moieties are converted to alanine and sulphoacetaldehyde by the transfer of amine residue to pyruvate. At last, sulfite is formed due to the addition of water, and after some electron transfer, the final product, hydrogen sulfide, is obtained (Schubert et al., 2017). This hydrogen sulfide moiety is toxic to various bacteria and even to the host cells, and in most cases, it results in inflammation (Ijssennagger et al., 2016). Some bacteria show molecular adaptation or resistance against bile salt. Among gram-negative bacteria, *Salmonella enterica*, *Escherichia coli*, and *Vibrio cholerae* have efflux pumps, which reduce the intracellular bile salt concentration through the efflux mechanism, followed by bile salt resistance (Schubert et al., 2017). According to some enzyme assays, the intracellular materials come out of the cells due to the formation of leakage. Scientists have gone through some electromagnetic studies that prove that the bile environment can make the cells shrink and drain out the intracellular materials. So, it is proven that after bile exposure, the membrane integrity gets disrupted, which can also be an antibacterial property of bile (Begley et al., 2005).

The steric orientation of the hydrophilic moieties of bile salt on one side facilitates the formation of micelles, which make bile salt symbiotic in nature in lipid digestion and solubilization (Espinal-Ruiz et al., 2016). However, this also determines the detergent nature of bile salts, which makes bile salts toxic to both host cells and microbiomes. Moreover, the hydrophobic bile salts are more toxic than the hydrophilic ones. The glycine and taurine-conjugated isoforms of bile acids are less hydrophobic than free bile acids. Further, the bile acid modification *via* deconjugation and dihydroxylation makes the bile acid more hydrophobic and more toxic for the human gut microbiota (Jia et al., 2018). But in the case of

rodents, bile salts have less hydrophobicity, which makes bile salts less toxic than human bile in a drastic way (Wahlström et al., 2017).

The gut microbiota has an immense impact on some metabolic disorders like diabetes, atherosclerosis, and other obesity-related disorders (Karlsson et al., 2013). The continuous interactions between host immune cells and the microbiota of the gut play the most important role in maintaining intestinal homeostasis and preventing inflammation (Gonçalves et al., 2018). The gut microbiota produces some metabolites to modulate the immune response by communicating with the immune cells (Goto & Kiyono, 2012). Studies revealed that intestinal antigen-presenting cells have the criteria to protect against infection, besides maintaining normal gut microbial flora (Suzuki et al., 2004). Studies have reported that the pro- and anti-inflammatory immune equilibrium in the gut is a condemnatory factor for balancing gut homeostasis, and the commensal microbial growth in the small intestine directly affects this system. The biological detergent activity of bile salts helps to control the commensal microbiomes and thus maintain intestinal homeostasis and immunity (Urdaneta & Casadesús, 2017). An enteric pathogen can respond to the enteric environment, and it can modify its genetic distribution to show virulent activity. As well as other environmental factors (pH, temperature, and osmolarity), bile also acts as an environmental signal that can control virulence. For example, bile suppresses the invasion of *Salmonella*; on the other hand, it enhances the *Shigella* invasion (Begley et al., 2005).

Role of gallbladder in our metabolism

The gallbladder is the storage place of bile acid, and when needed, it acts as a pacemaker for the secretion of bile acid (Qi et al., 2019). In some previous studies, it has been demonstrated that an increase in the concentration of bile acid can enhance the metabolism of glucose and lipids (Pathak et al., 2018). As we know, the farnesoid X receptor (FXR) is a bile acid receptor, which is a nuclear receptor and acts in the regulation of genes working on lipoprotein metabolism (Pineda Torra et al., 2003). Furthermore, FXR plays a crucial role in carbohydrate metabolism by regulating the genetic expression of phosphoenolpyruvate carboxykinase (PEPCK) (Stayrook et al., 2005). In this case, studies have shown that FXR agonists have a pivotal role in the regulation of the mRNA expression of hepatic PEPCK, resulting in glucose output from the hepatocytes, followed by carbohydrate metabolism (Fig. 3.4). This demonstrates that

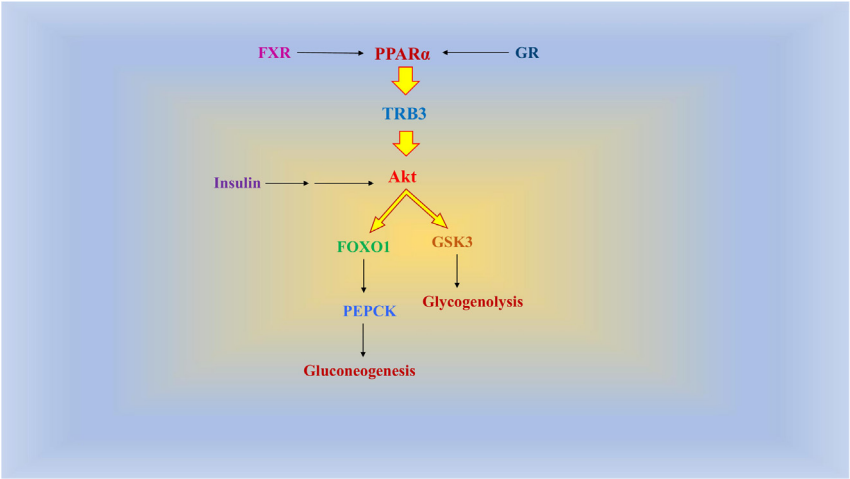

Figure 3.4 Estimated pathway of carbohydrate metabolism by FXR regulated pathway. Expression of PPAR gets enhanced because of the FXR mediated regulation. Later, TRB3 expression is increased and resulting into the binding of it to Akt. As a result, FOX1 and GSK3 phosphorylation gets stopped and followed by the enhancement of gluconeogenesis and glycogenolysis.

the bile acid receptor FXR has a connection with diabetes. Likewise, bile acid has the property to alter the genetic expression of PEPCK, glucose-6-phosphatase (G-6-Pase), and fructose-1,6-biphosphatase, resulting in the modification of gluconeogenesis (Stayrook et al., 2005). On the other hand, *fxr* knockout mice have demonstrated insulin resistance and further hyperglycemia (Li & Chiang, 2012). FXR also helps to regulate glucose-dependent insulin secretion as a result of its expression in the pancreatic β cells (Auti et al., 2022; Renga et al., 2010). The G-protein-coupled receptor TGR5 is a bile acid receptor present in the intestine, brown adipose tissues, white adipose tissues, and gallbladder (Pathak et al., 2018). Bile acid activates this TGR5, resulting in cellular activation of cAMP and PKA induction. This leads to increased energy expenditure and finally weight loss, and thus glucose homeostasis is maintained (Rajagopal et al., 2013; Stayrook et al., 2005).

The bile acid synthetic pathway is one of the major pathways for cholesterol catabolism (Russell, 2003). The enhancement of the catabolism of cholesterol in the intestine usually induces LDL receptor activity, resulting in LDL cholesterol uptake. This hepatic pathway decreases the peripheral cholesterol or cholesterol level in the blood (Luo et al., 2020). In contrast, some animal models have shown that hepatic bile synthesis has been reduced

by FXR receptor activation with its potent agonist, resulting in a drastic fall in serum cholesterol (Thomas et al., 2008). However, it has been clear for a long time that potent FXR agonists have been found to have an anti-atherosclerotic profile (Hageman et al., 2010). Similarly, a normal HDL level in serum represents a good sign for the transport of cholesterol to the liver, followed by decreased cholesterol saturation in bile. Paradoxically, the HDL metabolism under the influence of FXR is still unclear, as FXR acts as a hindrance in the production of apolipoprotein A1, the major component of HDL, resulting in a reduced concentration of HDL in the liver (Claudel et al., 2002). The genetic modulation of FXR-α by bile acid plays a core role in maintaining the homeostatic mechanisms of triglycerides. Further studies demonstrated that this FXR-α regulation is responsible for the expression of apolipoprotein C-II (APOC2), a coactivator of lipoprotein lipase, followed by a decrease in triglyceride levels in the serum (Kast et al., 2001). PPARγ coactivator 1α (PGC1α) has been found to coactivate FXR-α, which maintains energy homeostasis, and thus serum triglyceride levels also get reduced by the clearance of very low-density lipoprotein from the blood (Zhang et al., 2004).

Nonalcoholic fatty liver disease has become a life-threatening metabolic syndrome in the present scenario. If left untreated for a long time, it can even cause liver cirrhosis, gradually form hepatocellular carcinoma, and progress to liver failure. As FXR and TGR5 receptors of bile acid have been found to regulate energy homeostasis, maintain carbohydrate and lipid levels in the serum, and have an anti-inflammatory profile in the body, that is why both FXR and TGR5 receptor agonists can be a big therapeutic advancement for the treatment of nonalcoholic fatty liver disease and metabolic syndrome (Li et al., 2013).

Conclusion

Hence, it is clear that the biliary tree and gallbladder epithelium have a pioneering role in maintaining the body's innate and adaptive immunity. The gallbladder acts as the pacemaker for the secretion of bile acid. Previously, we discussed that numerous bile salt receptors in the plasma membrane and some nuclear receptors have frontier roles in the expression of bile salt actions. The detergent nature of bile salt has a toxic effect against microorganisms, and thus it maintains human immunity; however, it becomes toxic for the gut microbiota when it is in higher concentrations. On the other hand, the fernasoid X receptor occurs as an

inducing factor for gluconeogenesis and maintains carbohydrate homeostasis; besides, it reduces hypercholesterolemia. Both bile acid receptors, FXR and TGR5, have been found to reduce nonalcoholic fatty liver disease. Hence, further investigation on the metabolic effect of bile salt receptors must be carried out for the enhancement of the treatment quality of metabolic syndromes.

Conflict of interest

As per the authors, there is no conflict of interest.

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